

Development of an Embedded HPEC Telemetry Unit and Windshield HUD for the TUKZIE Rev 0 Platform



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Declaration

1. I know that plagiarism is wrong. Plagiarism is to use another's work and pretend that it is one's own.
2. I have used the IEEE convention for citation and referencing. Each contribution to, and quotation in, this report from the work(s) of other people has been attributed, and has been cited and referenced.
3. This report is my own work.
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Date: August 7, 2026

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Acknowledgments

Make relevant acknowledgements to people who have helped you complete or conduct this work, including sponsors or research funders.

Abstract

- Open the **Project Report Template.tex** file and carefully follow the comments (starting with %).
- Process the file with **pdflatex**, using other processors may need you to change some features such as graphics types.
- Note the files included in the **Project Report Template.tex** (with the .tex extension excluded). You can open these files separately and modify their contents or create new ones.
- Contact the latex manual for more features in your document such as equations, subfigures, footnotes, subscripts & superscripts, special characters etc.
- I recommend using the **kile** latex IDE, as it is simple to use.

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Chapter 1

Introduction

1.1 Background to the study

The global transition toward sustainable urban mobility has catalysed the development of lightweight, electric-drive vehicles capable of autonomous operation [1]. Within this context, the TUKZIE Flagship Programme seeks to engineer an energy-aware, secure vehicle platform for last-mile logistics and urban freight applications. The TUKZIE Rev 0, a 72V 2200W electric cargo trike, represents a testbed for integrating advanced telemetry and driver-assistance systems in resource-constrained environments.

Traditional vehicle telemetry systems have relied heavily on “Head-Down Displays” (HDDs) or post-processing of logged data, both of which present significant limitations. HDDs require drivers to divert their gaze from the road, contributing to cognitive overload and attention capture—a phenomenon where the driver’s focus shifts away from critical environmental cues [2, 3]. Post-processing delays maintenance interventions and prevents real-time decision-making.

Recent advances in High-Performance Embedded Computing (HPEC) have enabled edge-based signal processing in vehicular applications, allowing for real-time sensor fusion and on-vehicle analytics [4]. Simultaneously, Head-Up Display (HUD) technology has matured, offering a safer interface that projects critical information directly into the driver’s line of sight [5, 6]. However, the integration of these technologies—particularly for low-cost, three-wheeled electric vehicles in developing contexts—remains an open challenge.

This study addresses this gap by developing an integrated HPEC telemetry unit and windshield HUD for the TUKZIE Rev 0, enabling real-time ride-smoothness characterisation and driver feedback.

1.2 Objectives of this study

1.2.1 Problems to be investigated

Transportation agencies and fleet operators face significant challenges in monitoring road health and vehicle performance cost-effectively. Traditional profiling equipment is prohibitively expensive, with commercial pavement evaluation vehicles costing upwards of \$50,000–\$150,000 [7]. Standard in-car displays often contribute to cognitive overload and attention capture, where drivers divert their gaze from the road to process telemetry [8]. Furthermore, in resource-constrained environments, there is a gap for a modular, low-latency system capable of edge-based signal processing for ride characterisation while providing a safe, augmented-reality interface.

Specifically, this study investigates the following research questions:

1. How can a low-cost edge-processing system characterise ride-smoothness in real time using inertial sensors on a 72V electric cargo trike?
2. What is the optimal architectural approach for integrating a Head-Up Display with edge-based telemetry while minimising driver distraction?
3. What are the latency constraints for real-time data ingestion, edge processing, and 4G/LTE transmission to a central database?
4. How can true road roughness be isolated from nominal engine vibrations using statistical feature extraction?

1.2.2 Purpose of the study

The significance of investigating these problems lies in bridging the gap between expensive, proprietary road-profiling equipment and the need for cost-effective, real-time monitoring solutions. By leveraging High-Performance Embedded Computing and low-cost

inertial sensors, this study aims to provide a modular, low-latency system that delivers augmented-reality telemetry while ensuring driver safety.

The findings will directly inform the development of the TUKZIE Rev 0 platform and contribute to the broader field of sustainable urban mobility by demonstrating a replicable, open-architecture approach to vehicle telemetry. This work aligns with the UCT Radar Remote Sensing Group’s focus on advanced sensing and the TUKZIE Flagship Programme’s mandate for energy-aware, secure vehicle platforms.

Furthermore, this research addresses the practical needs of fleet operators in developing regions, where infrastructure monitoring budgets are constrained and traditional data collection methods are often infeasible. By enabling continuous, automated pavement condition assessment, the proposed system supports more efficient maintenance planning and improved road safety.

1.3 Scope and Limitations

1.3.1 Scope

This study focuses exclusively on the TUKZIE Rev 0, a 72V 2200W electric cargo trike manufactured by SA Motorcycles [9]. The system integrates the following components:

- A hardware module housing inertial measurement units (IMUs), air-quality sensors, and environmental sensors.
- A multithreaded software architecture for real-time sensor fusion, telemetry processing, and ride-smoothness indexing.
- An optical Head-Up Display subsystem projecting critical telemetry (speed, battery state of charge, and navigation) and safety alerts onto the trike’s windshield.
- A 4G/LTE communication link for central database synchronisation.

The project is aligned with the UCT Radar Remote Sensing Group’s focus on advanced sensing and specifically addresses the “ride-smoothness” mandate of TUKZIE by isolating true road roughness from nominal engine vibrations using statistical feature extraction.

The scope is limited to the TUKZIE Rev 0 platform and utilises low-cost inertial and environmental sensors processed at the edge using microcontroller and single-board computer platforms [7].

The software architecture follows a modular design, with separate threads for:

- High-frequency sensor polling (IMU at ≥ 100 Hz).
- Compute-heavy signal-processing algorithms (ride characterisation).
- Graphical HUD rendering.
- Networking stacks (MQTT/HTTP over 4G/LTE-M).

This separation ensures that no process blocks another [10].

1.3.2 Limitations

The following limitations are acknowledged:

- The study is limited to a single vehicle platform (TUKZIE Rev 0), which may affect generalisability to other vehicle types or configurations.
- Sensor calibration assumes nominal operating conditions; significant environmental variations (extreme temperatures and humidity) may affect accuracy.
- Edge-processing latency measurements are conducted in controlled test environments; real-world performance may vary with network conditions and electromagnetic interference.
- The HUD subsystem uses projected-display technology on the existing windshield, not full holographic or augmented-reality integration [5].
- Time constraints limit extensive field testing to urban routes only; rural and off-road conditions are not evaluated.
- The system is designed as a removable module to ensure that the vehicle can be quickly returned to a standard setup without obstructions [10].
- The 4G/LTE communication depends on network availability; data synchronisation may be delayed in areas with poor coverage.

1.4 Plan of development

This report is organised as follows:

Chapter 2: Literature Review—Reviews existing telemetry systems for electric vehicles, edge-computing architectures for vehicular applications, pavement condition-assessment methods, and Head-Up Display technologies. Key works in pavement roughness evaluation using inertial sensors [7, 11] and HUD design principles [5, 6, 12] are critically examined.

Chapter 3: System Design and Methodology—Presents the hardware architecture, sensor selection, and software framework for the integrated telemetry unit. The multithreaded architecture, sensor protocols (I2C, SPI, and UART), and edge-processing pipeline are detailed.

Chapter 4: Implementation—Details the fabrication process of the hardware module, HUD integration, and communication protocols. The PCB design, enclosure fabrication, and system assembly are described.

Chapter 5: Results and Discussion—Presents validation results for data ingestion accuracy, edge-processing latency analysis, ride-smoothness characterisation, and 4G/LTE data-synchronisation performance. Statistical analysis of sensor data and comparison with reference measurements are provided.

Chapter 6: Conclusions and Recommendations—Summarises the findings, evaluates the system against the stated objectives, and suggests future work, including potential integration of machine learning for predictive maintenance and extension to multi-vehicle fleet monitoring.

Chapter 2

Literature Review

Once upon a time engineers and researchers believed... In this area of research, they used the following methods... [14]

Write this section first as it will take you the longest. I suggest you start writing this as soon as you have done your initial research at the beginning of your project. You can then return to it once you have completed your work to edit and adjust it.

A literature review forms the theoretical basis of your project. You need to read a large number of journal papers, sections in books, technical reports etc. relevant to your work at the start of project. This will give you a good idea of the field of research.

When writing your review start of with the general concepts and move to the more specific aspects explaining the necessary theory as you go. This section is NOT a copy and paste from others work or a rewrite-but-change-one-word section. I suggest you read all your material, and then put it down and write this section, referring back to the work only when you need to check something.

See your PCS textbook for more details on how to write a literature review.

If you include a figure or a table in your text please see the example in Fig. 2.1 as to how to caption it. Please make sure that all text in your figures is readable and that you reference your figures if they are from another source.

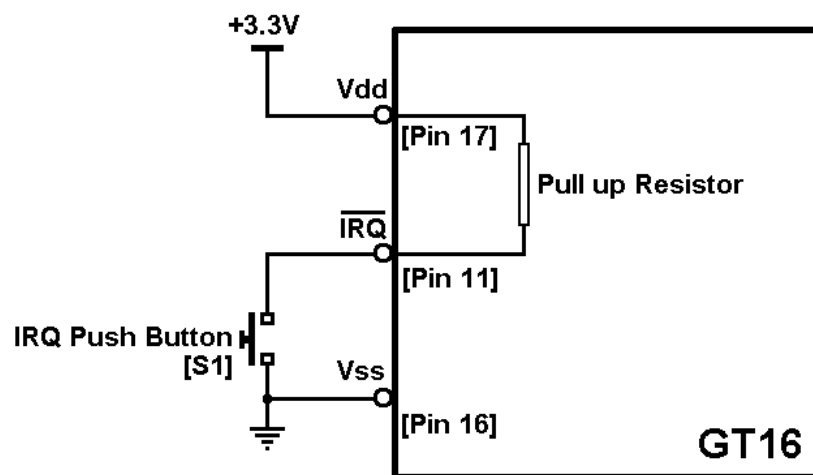


Figure 2.1: A block diagram illustrating the connections to the IRQ pin on the MCS08GT16A microcontroller (Please note that your headings should be short descriptions of what is in the diagram not simply the figure title)

Chapter 3

Methodology

This is what I did to test and confirm my hypothesis.

You may want to split this chapter into sub chapters depending on your design. I suggest you change the title to something more specific to your project.

This is where you describe your design process in detail, from component/device selection to actual design implementation, to how you tested your system. Remember detail is important in technical writing. Do not just write I used a computer give the computer specifications or the oscilloscopes part number. Describe the system in enough detail so that someone else can replicate your design as well as your testing methodology.

If you use or design code for your system, represent it as flow diagrams in text.

IMPORTANT: Include a motivation for your selection of an appropriate technique, or engineering or IT tool to solve your project problem. Discuss any limitations if appropriate.

Chapter 4

Results

These are the results I found from my investigation.

Present your results in a suitable format using tables and graphs where necessary. Remember to refer to them in text and caption them properly.

4.1 Simulation Results

4.2 Experimental Results

Chapter 5

Discussion

Here is what the results mean and how they tie to existing literature...

Discuss the relevance of your results and how they fit into the theoretical work you described in your literature review.

Chapter 6

Conclusions

These are the conclusions from the investigation and how the investigation changes things in this field or contributes to current knowledge...

Draw suitable and intelligent conclusions from your results and subsequent discussion.

Chapter 7

Recommendations

Make sensible recommendations for further work.

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Appendix A

Additional Files and Schematics

Add any information here that you would like to have in your project but is not necessary in the main text. Remember to refer to it in the main text. Separate your appendices based on what they are for example. Equation derivations in Appendix A and code in Appendix B etc.

IMPORTANT: Appendix A (see the table below) should provide a summary of how you have met the GAs associated with the course, and where in the report the evidence can be found.

If you have used AI in writing your report, you need to provide details in one of the appendices of how you used it.

If appropriate, in a subsequent appendix, provide a link to any GitHub repository where the code and additional materials for your project can be found.

A.1 Declaration of Artificial Intelligence Use

Artificial intelligence tools were used to assist with clarifying the organisation of this report, improving the alignment and consistency of its presentation, refining wording, and checking LaTeX formatting. All AI-assisted suggestions were reviewed by the author. The author remains responsible for the report's technical content, analysis, interpretations, and final editorial decisions.

APPENDIX A. ADDITIONAL FILES AND SCHEMATICS

GA	Requirement	Justification and section in the report
1	Problem-solving	Description here
4	Investigations, experiments and data analysis	Description here
5	Use of engineering tools	Description here
6	Professional and technical communication (Long report)	Description here
8	Individual work	Description here
9	Independent learning ability	Description here

Appendix B

Addenda

B.1 Ethics Forms